

Gate 2 Record

GATE 2 — SENSING VERIFICATION RECORD

CIP-BTSB-BT-001 · Supporting Record G2

CIP — Compute Infrastructure Platform · S. Mohammed Idreesh · Chennai Test date: 24 August 2026 · Raw data: `gate2_sensing_check_20260824.csv`

[PRE-GEN-0 MECHANISM DEMONSTRATION — NOT A PRODUCT VALIDATION]

1. PURPOSE

To verify, before any heat is applied, that the two-channel thermocouple system reads correctly, reads reliably, and to characterise the instrument error that must be subtracted from the test results.

This record is referenced by Section 4.3 of the pre-registered test report.

2. CONFIGURATION AS BUILT

Item	Detail
Controller	Arduino Uno R3, ATmega328P, powered from laptop USB
Amplifiers	2 × MAX6675, module pin order GND VCC SCK CS S0
Sensors	2 × K-type thermocouple, M6 screw tip, braided cable
Sketch	BTSB_bench_logger.ino , 4,882 bytes, 1 Hz CSV output
Library	Adafruit MAX6675 v1.1.2
Serial	9600 baud, COM7

2.1 Pin map

SHARED (both modules)	DISTINCT
VCC → Arduino 5V	module #1 CS → Arduino D7
GND → Arduino GND	module #2 CS → Arduino D8
SCK → Arduino D6	
S0 → Arduino D5	

Shared lines are bridged on a solderless breadboard: four junction columns, one per shared signal, each carrying one wire in from the Arduino and one wire out to each module. Chip-select lines run point to point and do not touch the breadboard. Fourteen conductors total.

2.2 Thermocouple termination

Probe leads terminate in the module’s screw terminal block, red to `+`, blue to `-`. **Crimped spade lugs were removed and replaced with bare stripped conductor** — see Section 6.

3. METHOD

Both probes were left free in still room air, at the same height, not touching any surface or each other. The system logged continuously at 1 Hz with no intervention.

With both sensors at the same true temperature, any indicated difference is instrument error. That difference is the quantity this test measures.

4. RESULTS

4.1 Reliability

Duration	14.0 minutes (0.3 s → 837.5 s)
Valid readings	836
NaN / sensor-fault rows	0
Fault rate	0.00 %

4.2 Channel statistics

	Mean °C	SD °C	Min	Max
TC1	34.017	0.282	33.00	34.75
TC2	34.646	0.303	33.75	35.50

4.3 Indicated channel difference in still air

T1 - T2	mean	= -0.629 °C
	median	= -0.750 °C
	SD	= 0.363 °C
	range	= -1.75 to +0.50 °C

THIS IS NOT AN INSTRUMENT OFFSET, AND NO CORRECTION IS DERIVED FROM IT.

An earlier revision of this record interpreted -0.63 °C as an instrument zero offset and directed that +0.63 °C be added to every measured difference. **That interpretation was wrong and the direction is withdrawn** — see §4.6 and §6.2.

The probes were separated in space during this run. The figure above therefore contains an unknown mixture of instrument offset and genuine spatial temperature difference within the room, and the two cannot be separated from this dataset.

4.4 Common-mode rejection

Room temperature rose during the run. Both channels tracked it:

	TC1	TC2	T1 - T2
First 100 s	33.88	34.44	-0.565
Last 100 s	34.29	34.87	-0.574
Change	+0.41	+0.42	-0.009

Ambient drifted 0.41 °C. The measured difference between the channels changed by 0.009 °C.

This is the most important result in the record. The bench test measures a *difference* between two plates, and the analysis compares *differences between runs*. A pair of sensors whose offset shifts with ambient would corrupt that comparison invisibly. These do not: common-mode drift is rejected by a factor of roughly 45.

4.5 Resolution

Eleven distinct values were recorded across both channels, with a uniform step of 0.25 °C — matching the MAX6675 12-bit specification. Resolution is confirmed as 0.25 °C; observed noise is approximately one quantisation step.

4.6 Elevated-temperature verification — boiling-water reference

The still-air run above establishes behaviour at 34 °C only. The bench test operates between 60 and 110 °C. Two immersion checks were therefore performed against boiling water. Chennai is at sea level, so the reference is 100.0 °C.

Check 1 — TC2 fully immersed in vigorously boiling water, TC1 shallower.

Phase	n	TC1	TC2	T1 - T2
Air, before immersion, probes together	8	33.81	33.84	-0.03
Boiling plateau, 40-70 s	30	91.48	99.47	-7.99
Cooling, both out of water	9	77.86	77.08	+0.78

TC2 peak: 101.00 °C against a 100.0 °C reference. **TC2 is verified at the boiling point to within +1.0 °C**, comfortably inside the ±2.5 °C Class 2 tolerance.

The 7.99 °C gap at the plateau is an **immersion-depth artefact, not a span error**. The proof is in the cooling phase: once both probes were removed from the water the readings converged and then **crossed**, with TC1 ending 0.78 °C *above* TC2. A genuine span error in TC1 would have read low at every elevated temperature and could not reverse sign.

Check 2 — both probe tips held together, deep, in the same water.

Window	n	TC1	TC2	T1 - T2	SD
Settled, 23-61 s	38	93.04	93.77	-0.73	0.81
Best, 27-45 s	18	93.72	94.50	-0.78	0.81

Peaks: TC1 95.25, TC2 96.25. Within the settled window, 22 of 38 sample pairs agreed to within 0.5 °C.

The verification chain is therefore: TC2 against the 100 °C boiling reference in Check 1, and TC1 chained to TC2 at 93-95 °C in Check 2. Both channels are verified across the operating range of the test.

And the decisive result for §4.3: with both probes held together in air the indicated difference was **−0.03 °C**, not −0.63 °C. Two probes at a genuinely identical temperature agree to well inside one quantisation step. The −0.63 °C seen in still air was therefore predominantly a **real spatial temperature difference between two points in the room**, not an instrument characteristic.

5. INTERPRETATION

Quantity	Value	Note
Absolute accuracy	±2.5 °C	K-type Class 2 tolerance
Resolution	0.25 °C	confirmed by measurement, §4.5
TC2 at boiling reference	101.0 vs 100.0 °C	+1.0 °C , §4.6
TC1 chained to TC2 at 93-95 °C	−0.73 ± 0.81 °C	§4.6
Channel agreement, tips together in air	−0.03 °C	§4.6
Common-mode rejection	±0.01 °C over 0.4 °C ambient	§4.4
Fault rate	0.00 % over 14 min	§4.1
Fixed offset correction applied	NONE	see below

Absolute temperatures from this system should be read with the ±2.5 °C caveat. Differences between runs should not — any systematic sensor offset is common to all runs and cancels in the comparison, and §4.4 demonstrates that the channel relationship is stable against ambient drift.

5.1 Offset handling in service — the operative instruction

No fixed correction is applied to any run in this programme.

The channel difference is **re-measured at the start of every run**, with both probes mounted on their plates and the fixture at ambient, and that per-run baseline is recorded on the field log sheet and in §10.1 of CIP-BTSB-BT-001.

This supersedes the withdrawn +0.63 °C instruction. Measuring the baseline in the as-built configuration immediately before each run captures both instrument offset and fixture-specific spatial difference together, which is the quantity that actually matters, and it does so without relying on an assumption that either component is constant.

6. FAULTS FOUND AND CORRECTED

Recorded in full because they are part of the instrument’s history.

6.1 Intermittent open circuit at the probe termination

Symptom. During initial single-channel testing, **NaN** sensor-fault rows appeared intermittently — typically two to five consecutive faults every few seconds, resolving without intervention.

Diagnosis. The fault correlated with physical movement of the probe cable. Holding the assembly still produced clean output; releasing it produced faults. This indicated an intermittent open circuit rather than an electrical or software problem.

Cause. The probe leads were terminated with crimped flat spade lugs inserted into a screw terminal block designed for round bare conductor. Contact was made only at the blade edges, and the weight of the braided cable was sufficient to break it.

Correction. Spade lugs removed. Conductors stripped, twisted and clamped directly. Cable strain-relieved so that its weight is not carried by the terminal.

Verification. 836 consecutive readings over 14 minutes with zero faults, as recorded in Section 4.1.

6.2 Misinterpretation of the still-air channel difference — analysis error

Error. The first issue of this record interpreted the −0.63 °C mean difference of §4.3 as an instrument zero offset, and directed the operator to add +0.63 °C to every measured T1 – T2 value in the test programme.

Detection. The boiling-water checks of §4.6 included an interval with both probe tips held physically together in still air. The indicated difference in that interval was **−0.03 °C** — twenty times smaller, and well inside one quantisation step.

Cause of the error. During the still-air run the probes were deliberately separated, at the same height but not touching. That separation was intended to avoid thermal coupling between them. Its consequence was that any genuine spatial temperature difference in the room was indistinguishable from instrument offset in the resulting data. The measurement could not support the conclusion drawn from it.

Correction. The +0.63 °C instruction is withdrawn in full. **It was never applied to any run**, as it was withdrawn before the first heated run. The replacement procedure is in §5.1: re-measure the baseline before every run, in the as-built configuration.

Consequence had it not been caught. The primary endpoint of the test is a difference between plates. Adding a fixed +0.63 °C to every reading would have displaced every reported temperature difference by that amount, in the same direction, in all three runs — and because it would have been applied consistently it would have looked entirely self-consistent in the results.

7. OUTSTANDING BEFORE FIRST HEATED RUN

- [✓] Boiling-water span check — COMPLETED, §4.6
TC2 verified at 101.0 °C against a 100.0 °C reference
TC1 chained to TC2 at 93-95 °C
- [✓] Probe assignment fixed in firmware and physically labelled
MAX6675 #1, CS on D7 = TC1 = HOT plate
MAX6675 #2, CS on D8 = TC2 = COLD plate
- [] Per-run baseline difference measured immediately before each run, with both probes mounted on the plates (§5.1)
– this is a per-run operating step, not a gate condition

8. GATE STATUS

GATE 2 — PASSED.

Reliability, resolution, common-mode rejection, fault detection and elevated-temperature span all verified. The provisional condition on the boiling-water check is discharged by §4.6.

No fixed offset correction is applied. Per-run baseline measurement per §5.1.

Record G2 · First issued 24 August 2026 · Revised 25 August 2026 to add §4.6 elevated-temperature verification, to withdraw the +0.63 °C correction (§6.2), and to discharge the provisional gate condition. Raw data published alongside. Supports CIP-BTSB-BT-001. All build-manual performance figures remain [GEN-0 VALIDATION REQUIRED].